

Quantitative CTA guidelines

Practice guidelines

Standards for quantitative assessments by coronary computed tomography angiography (CCTA)

An expert consensus document of the society of cardiovascular computed tomography (SCCT)

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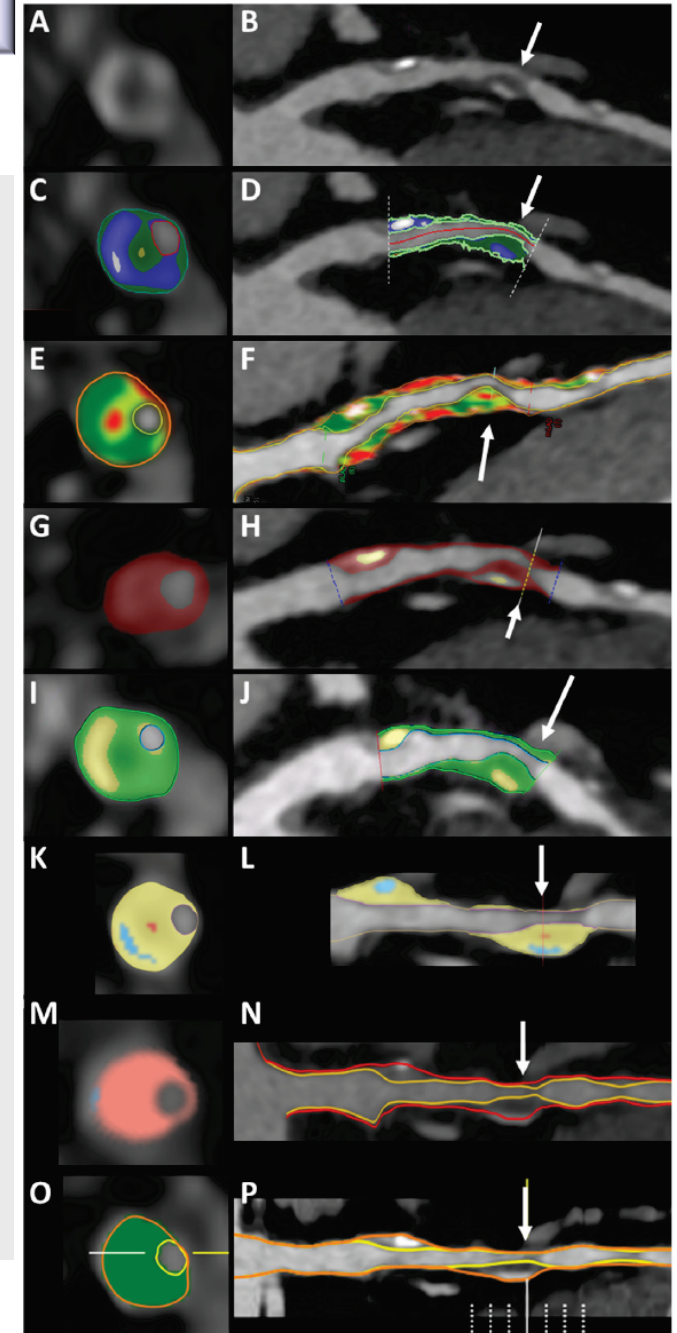
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Table 3

Effect of selected CT parameter modifications on quantitative plaque assessment.

CT parameter modification	Effect on image quality	Effect on plaque quantification
↓ Tube voltage	↑ Lumen attenuation ↑ Attenuation values for all plaque components	↑ Calcified plaque volume ↓ Low-attenuation plaque volume
↑ Tube current	↓ Image noise ↑ Overall image quality	↑ Homogeneity of plaque attenuation values ↑ Reproducibility
↑ Intra-coronary iodine concentration	↑ Lumen attenuation ↑ Attenuation values for all plaque components	↑ Calcified plaque volume ↓ Low-attenuation plaque volume
↑ Reconstruction kernel sharpness	↑ Lumen contour sharpness ↑ Image noise	↓ Homogeneity of plaque attenuation values
↑ Iterative reconstruction strength	↓ Image noise	Small effects on attenuation values and plaque size in some studies

Effect of isolated modifications and assumption of fixed HU-thresholds for segmentation of inner and outer vessel boundaries and plaque components.

Details matter

- Software ideally should consider the effect of x-ray tube voltage, tube current, luminal contrast density, noise and reconstruction parameters on plaque output and lumen/wall segmentation – most currently do not
- Optimal to use the clinical-read CTA for plaque quantification from spectral CT

